



NeoStats

AI-Powered Credit Risk Intelligence Platform

Intelligence. Innovation. Impact.

LightGBM

ML Model

SHAP

Explainability

Groq LLaMA

Talk-to-Data

Business Problem

- Faster, more accurate credit decisions needed
- Explainability required for regulatory compliance
- Analysts need plain-English data exploration
- ~8% default rate with heavy class imbalance

Platform Architecture

Streamlit UI

5 Sections

LightGBM + SHAP

ML & Explainability

Groq NL→SQL

LLaMA 3.3 70B

SQLite Database

From CSV on init

EXPLORATORY DATA ANALYSIS

50,000

Applicants

8.05%

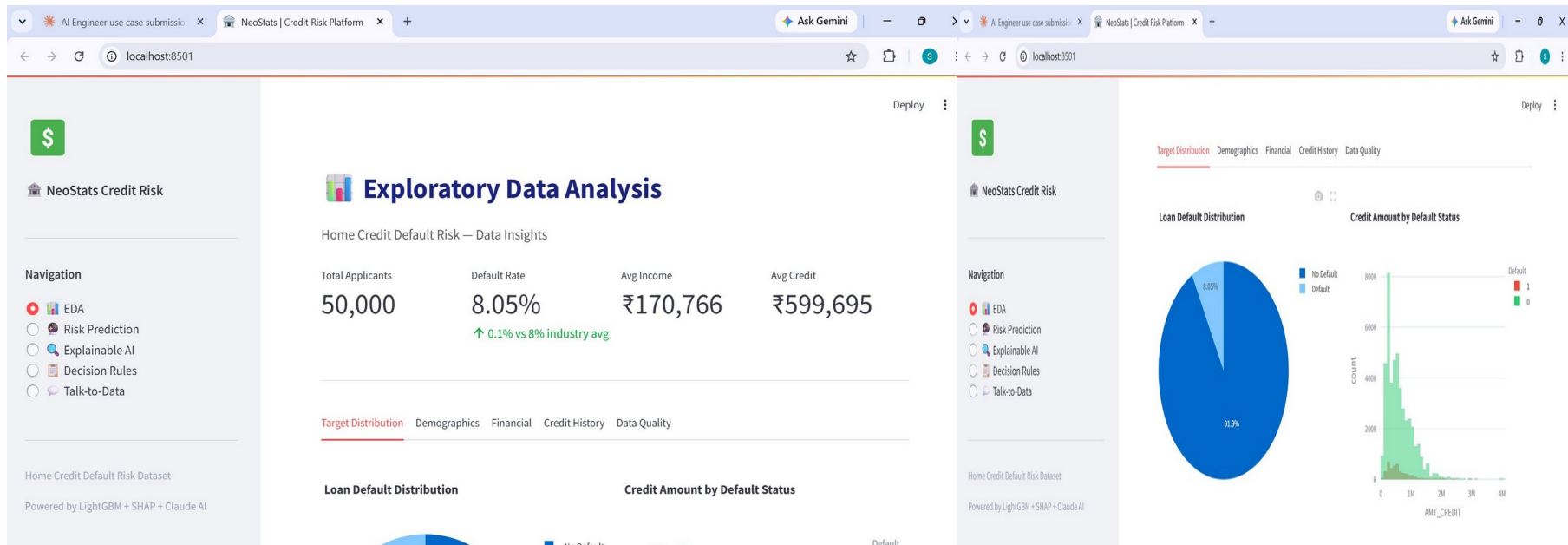
Default Rate

₹1.71L

Avg Income

₹5.99L

Avg Credit



💡 8% imbalance → scale_pos_weight

💡 Younger (20-35) = higher default

💡 EXT_SOURCE 2&3 = top predictors

CREDIT RISK PREDICTION

AI Engineer use case submission: x NeoStats | Credit Risk Platform x + Ask Gemini

localhost:8501

Deploy

NeoStats Credit Risk

Navigation

- EDA
- Risk Prediction**
- Explainable AI
- Decision Rules
- Talk-to-Data

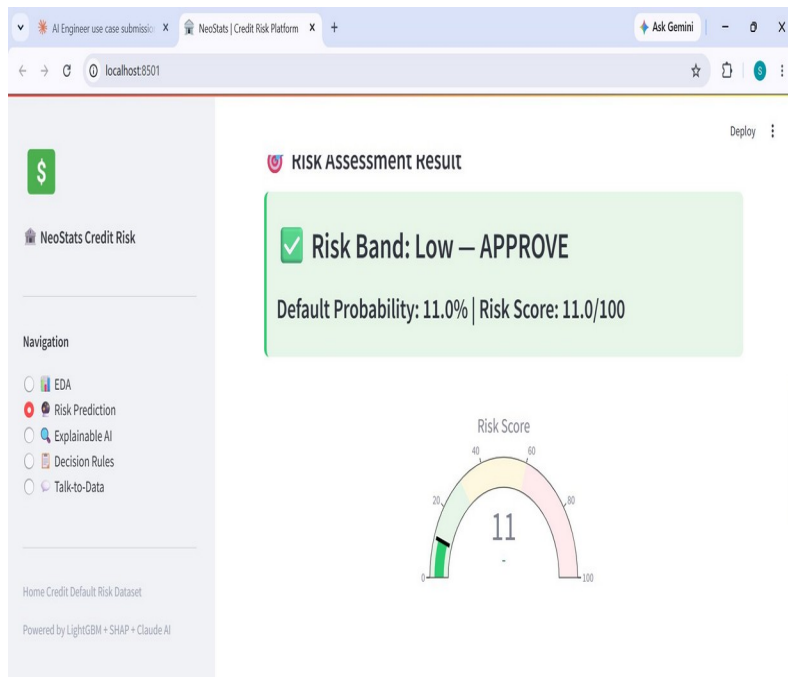
Home Credit Default Risk Dataset

Powered by LightGBM + SHAP + Claude AI

Applicant Information

Personal Details	Financial Information	Credit Scores & History
Gender: M	Annual Income (₹): 200000	External Score 1: 0.50
Age (years): 35	Loan Amount (₹): 500000	External Score 2: 0.50
Number of Children: 2	Monthly Annuity (₹): 25000	External Score 3: 0.50
Own Car?: Y	Goods Price (₹): 450000	Housing Type: House / apartment
Own Property?:	Income Type: Working	

Division Risk Dataset



⚡ Model

LightGBM

📊 ROC-AUC

0.75–0.77

🎯 PR-AUC

0.35–0.45

🛠️ Imbalance Fix

scale_pos_weight

Output: Risk Band (Low / Medium / High) + Default Probability + Risk Score (0–100) + SHAP values

The screenshot shows a web browser window with the URL `localhost:8501`. The page title is "NeoStats Credit Risk". On the left is a navigation sidebar with a green dollar sign icon and the text "NeoStats Credit Risk". Below this is a "Navigation" section with five radio buttons: "EDA", "Risk Prediction", "Explainable AI", "Decision Rules" (which is selected), and "Talk-to-Data". At the bottom of the sidebar, it says "Home Credit Default Risk Dataset" and "Powered by LightGBM + SHAP + Claude AI". The main content area displays a list of eight decision rules, each in a white box with a colored icon and a dropdown arrow. The rules are: R2: Low External Credit Score (red icon), R3: Young Applicant + Unemployed (red icon), R4: Positive External Score Combination (green icon), R5: High Annuity-to-Income Ratio (yellow icon), R6: Multiple Previous Refusals (yellow icon), R7: High Region Risk Rating (yellow icon), and R8: Stable Long-term Employment (green icon). A "Deploy" button with a three-dot menu is at the top right of the rules list.

Deploy

- R2: Low External Credit Score
- R3: Young Applicant + Unemployed
- R4: Positive External Score Combination
- R5: High Annuity-to-Income Ratio
- R6: Multiple Previous Refusals
- R7: High Region Risk Rating
- R8: Stable Long-term Employment

8 ML-Derived Rules

R1

Credit-to-Income > 5×

High

R2

EXT_SOURCE_2 < 0.35

High

R3

Young + Unemployed

High

R4

Triple High Ext Scores

Low

R5

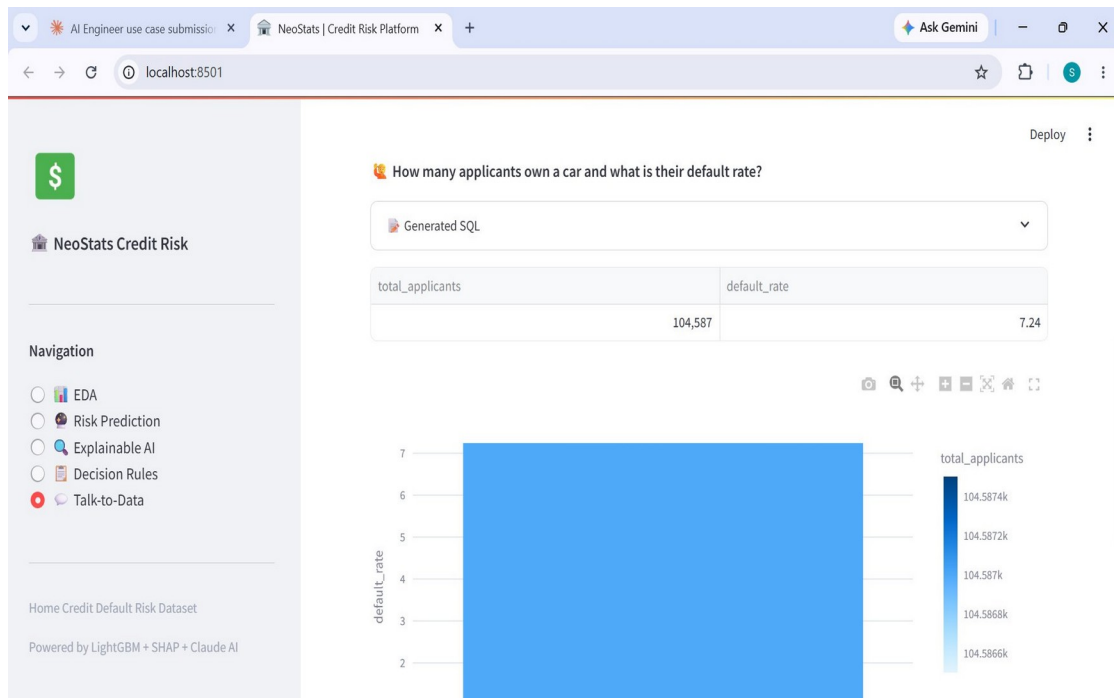
Annuity > 25% Income

Med

R6

2+ Prior Refusals

Med



How it works

- 1 User types question**
Plain English query
- 2 Groq LLaMA 3.3 70B**
Generates SQL query
- 3 SQL Validation**
Safety + syntax check
- 4 SQLite Execution**
Runs on real data
- 5 Insight Generation**
Human-readable answer

Tech Stack & Summary



Multi-section UI



ML + SHAP



NL→SQL Agent



Embedded DB



Interactive Charts



One-command deploy

Full-stack AI solution: EDA → ML → XAI → NL Chatbot → Docker Deployment